Probabilistic Hierarchical Clustering

- Algorithmic hierarchical clustering
  - Nontrivial to choose a good distance measure
  - Hard to handle missing attribute values
  - Optimization goal not clear: heuristic, local search
- Probabilistic hierarchical clustering
  - Use probabilistic models to measure distances between clusters
  - Generative model: Regard the set of data objects to be clustered as a sample of the underlying data generation mechanism to be analyzed
  - Easy to understand, same efficiency as algorithmic agglomerative clustering method, can handle partially observed data
- In practice, assume the generative models adopt common distribution functions, e.g., Gaussian distribution or Bernoulli distribution, governed by parameters

Generative Model

- Given a set of 1-D points X = {x1, …, xn} for clustering analysis & assuming they are generated by a Gaussian distribution $N(\mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$
- The probability that a point xi ∈ X is generated by the model:

$$P(x_i|\mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x_i-\mu)^2}{2\sigma^2}}$$

- The likelihood that a data set X is generated by the model:

$$L(N(\mu, \sigma^2): X) = P(X|\mu, \sigma^2) = \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x_i-\mu)^2}{2\sigma^2}}$$

- The task of learning the generative model: find the parameters $N(\mu_0, \sigma_0^2) = \arg\max\{L(N(\mu, \sigma^2): X)\}$

A Probabilistic Hierarchical Clustering Algorithm

- For a set of objects partitioned into m clusters C1, . . . ,Cm, the quality can be measured by $Q(\{C_1,\ldots,C_m\}) = \prod_{i=1}^{m} P(C_i)$, where P() is the maximum likelihood
- If we merge two clusters C_{j1} and C_{j2} into a cluster $C_{j1} \cup C_{j2}$, the change in quality of the overall clustering is

$$Q\left(\left(\{C_1,\ldots,C_m\} \setminus \{C_{j1}, C_{j2}\}\right) \cup \{C_{j1} \cup C_{j2}\}\right) - Q(\{C_1,\ldots,C_m\})$$
$$= \prod_{i=1}^{m} P(C_i)\left(\frac{P(C_{j1} \cup C_{j2})}{P(C_{j1})P(C_{j2})} - 1\right)$$

- Distance between clusters C_i and C_j $dist(C_i, C_j) = -\log \frac{P(C_i \cup C_j)}{P(C_i)P(C_j)}$
- If $dist(C_i, C_j) < 0$, merge C_i and C_j

Example